

Total No. of Questions : 6]

SEAT No. :

PC-15

[Total No. of Pages : 2

[6360]-15

T.E. (Civil)

**HYDROLOGY AND WATER RESOURCES
ENGINEERING**

(2019 Pattern) (Semester - I) (301001) (Insem)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve Q.No.1 or 2, 3 or 4, 5 or 6.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Use of logarithmic table, slide rule and electronic pocket calculator are allowed.
- 5) Assume suitable data if necessary, stating it clearly.

- Q1)** a) Explain applications of hydrology. [5]
b) Explain National Institute of Hydrology (NIH). [5]

OR

- Q2)** a) Explain drizzle form and rain form of precipitation. [4]
b) What is infiltration capacity; explain any two factors affecting infiltration capacity. [3]
c) What are the different methods of measuring evaporation and draw sketch of class A evaporation pan. [3]

- Q3)** a) Explain working of symphonic rainguage with neat sketch. [5]
b) Explain frontal and orographic precipitation. [5]

OR

- Q4)** a) Explain BINNI'S method & BARLOW Tables for runoff estimation. [6]
b) State and explain factors affecting runoff. [4]

P.T.O.

- Q5) a)** State the assumptions made in Unit Hydrograph theory. [3]
- b)** Given below are ordinates 6-h unit hydrograph for a catchment. Calculate the ordinates of the DRH due to a rainfall of 3.5 cm occurring in 6 hours. [7]

Time (h)	0	3	6	9	12	15	18	24
UH Ordinates m ³ /s	0	25	50	85	125	160	185	160
Time (h)	30	36	42	48	54	60	69	
UH Ordinates m ³ /s	110	60	36	25	16	8	0	

OR

- Q6) a)** Explain velocity area method for stream gauging. Draw neat sketch. [7]
- b)** Explain components of typical hydrograph. [3]

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